SUMANTH MANOHAR

Data Scientist | Machine Learning Engineer

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SUMMARY

Data Scientist and ML Engineer with 3+ years shipping production ML and LLM systems at Walmart Global Tech and Maxgen Technologies. Builds forecasting and NLP models, RAG systems, and the evaluation and monitoring that keep them reliable after launch. MS in Data Science, University of Arizona.

SKILLS

ML & Statistics: Forecasting, classification, recommendation systems, XGBoost, LightGBM, PyTorch, TensorFlow, A/B testing, hypothesis testing

NLP & GenAI: LLMs, RAG, vector databases (Pinecone, Milvus), prompt engineering, LLM evaluation, Transformers (HuggingFace), spaCy

Data & MLOps: Python, SQL, R, Apache Spark, Databricks, Airflow, Snowflake, MLflow, Docker, Kubernetes, FastAPI, CI/CD

Cloud: AWS (SageMaker, Lambda), GCP (Vertex AI)

EXPERIENCE

Data Scientist | Walmart Global Tech, USA **May 2025 – Present**

- Improved demand-forecast accuracy 18% by building end-to-end forecasting pipelines on 500K+ daily supply-chain records with Python, Apache Spark, and GCP Vertex AI.
- Cut analyst research time 40% and improved answer precision 25% by building an LLM-powered RAG knowledge retrieval system on vector databases.
- Reduced unplanned model retraining 35% by automating production monitoring with drift detection, performance tracking, and MLflow registry workflows.
- Lowered compute costs 22% with no latency loss at peak load by containerizing ML inference on Docker and Kubernetes with horizontal auto-scaling.

Data Scientist | Maxgen Technologies, India **Feb 2021 – Jun 2023**

- Lifted user conversion 12% by designing A/B testing frameworks for personalized recommendation models and iterating on live user behavior.
- Raised extraction precision 22% and eliminated 60+ hours/month of manual review by productionizing transformer NLP pipelines for classification, NER, and extraction.
- Improved classification accuracy 28% on high-volume datasets with XGBoost, LightGBM, and TensorFlow, using cross-validation, tuning, and class-imbalance handling.
- Cut new-model lead time by two weeks and improved deployment reproducibility 30% by building a shared feature store and deploying models on AWS SageMaker with Docker and Terraform.
- Helped drive a 20% increase in cross-functional AI adoption by adding SHAP/LIME explanations that made model decisions clear to non-technical stakeholders.

PROJECTS

LLM Evaluation & Regression Pipeline | *Python, Anthropic/OpenAI APIs, GitHub Actions*

- Built CI-gated evaluation that scores every prompt change against a hand-curated golden dataset with LLM-as-judge rubrics and stress cases, blocking releases that drop accuracy beyond set thresholds.

AI Pipeline Forensics | *Python, FastAPI, Streamlit*

- Built a root-cause analyzer for multi-step LLM pipelines that traces each step, pinpoints where silent failures begin, and turns every confirmed failure into a regression test.

LLM Cost Autopilot | *Python, FastAPI, scikit-learn*

- Cut LLM API cost 59% on synthetic traffic with a gateway that routes each request to the cheapest adequate Claude or GPT tier in under 5ms, escalating automatically when quality drops.

Cloud Forensics Using LLMs (research paper in progress) | *Python, RAG, AWS*

- Cut investigation time 60% with an LLM and RAG pipeline that triages 500+ AWS CloudTrail logs daily.

EDUCATION

Master of Science in Data Science | University of Arizona, USA **Dec 2025**

Bachelor of Engineering in Computer Science | Dayananda Sagar College of Engineering, India **2020 – 2024**

CERTIFICATIONS

Foundation Certificate in Programming and Data Science, IIT Madras (Online)